



EPC

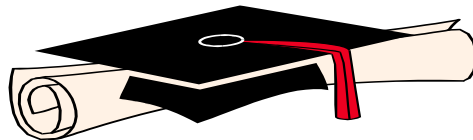
Elementary Programming with C

Semester: 1

The Exam Code: 02

Duration: 90 minutes

Total Marks: 20



Practical Examination Paper

Do not write on this question paper and return it to the Invigilator after the examination.

Question 1:

[4]

Write a program that asks the user to input the ages of 5 students. Use an array to store the age values. Ensure that the age values must be greater than 5 and less than 18. If this condition is not satisfied, ask the user to input the age again. After collecting the ages, display the minimum age, maximum age, and the average age of all the students.

Sample Output:

```
Please enter the ages of 5 students:
Student 1: 12
Student 2: 7
Student 3: 20
Age must be greater than 5 and less than 18.
Please enter the age for Student 3 again: 14
Student 4: 9
Student 5: 16
Minimum age:
7 Maximum age:
16 Average age: 11.6
```

Question 2:

[4]

Write a program that accepts a string and uses a pointer to determine whether the string is a palindrome, meaning it reads the same forward and backward.

Sample Output:

```
Enter a string: racecar
The string is a palindrome.
Enter a string: abcd
The string is not a palindrome.
```

Question 3:

[12]

Create a Employee management system program in C

Note that you must use the pointer concept to do this task.

Create a data structure following the specifications

[1]

Data type	Field	Description
char	name[20]	Name of employee
char	country[20]	Country of employee
int	birthyear	Birth year of employee
float	salary	Salary of employee

Create menu based program as follows:

[2]

Employee management system
1. Input Employee 2. Display Employee Descending 3. Save file 4. Analyze 5. exit

Input list Employees

[2]

When the user selects the option “1. Input employee” from the menu

do the followings:

- Ask the user to enter the total number of employees
- Input information for each employee in the list
- Validate data input by the following rules:

Data anomalies	Message
$60 \geq \text{current_year} - \text{birthyear} \geq 18$	Employee age must less than 18

	years old and greater than 6 years old
100 <= salary <= 500	Employee salary must less than 500 and greater than 100

- If data input is invalid, ask the user to re-enter.

Display employee: [2]

When the user selects the option “2. Display Employee” from the menu

- Sort employees by name in ascending order.
- Displays details of each Employee in the list after the above re-arrangement.

Save to file: [2]

- When the user selects the option “3. Save file” from the menu, do the followings:
- Ask the user to enter a file name.
- Create the file in binary mode.
- If the program cannot open a file, notify the user about that error and return to end this function.
- Write data from the list into the file.
- Notify the user that data has been saved successfully.
- Close the file.

Analyze: [2]

When user selects the option “4. Analyze” from the menu :

- Calculate how many employees are there from a specific country.

- Display the statistics results. For example:

Sample data

John Doe	hanoi		1995	250
Alice Johnson	hanoi		1990	350
Michael Smith	hcm		1992	300
Sara Brown	hanoi		1988	200
David Lee	hcm		1995	450

3 employees from hanoi
2 employees from hcm

Exit:

[1]

When the user selects the option “5. Exit” then exit the program.